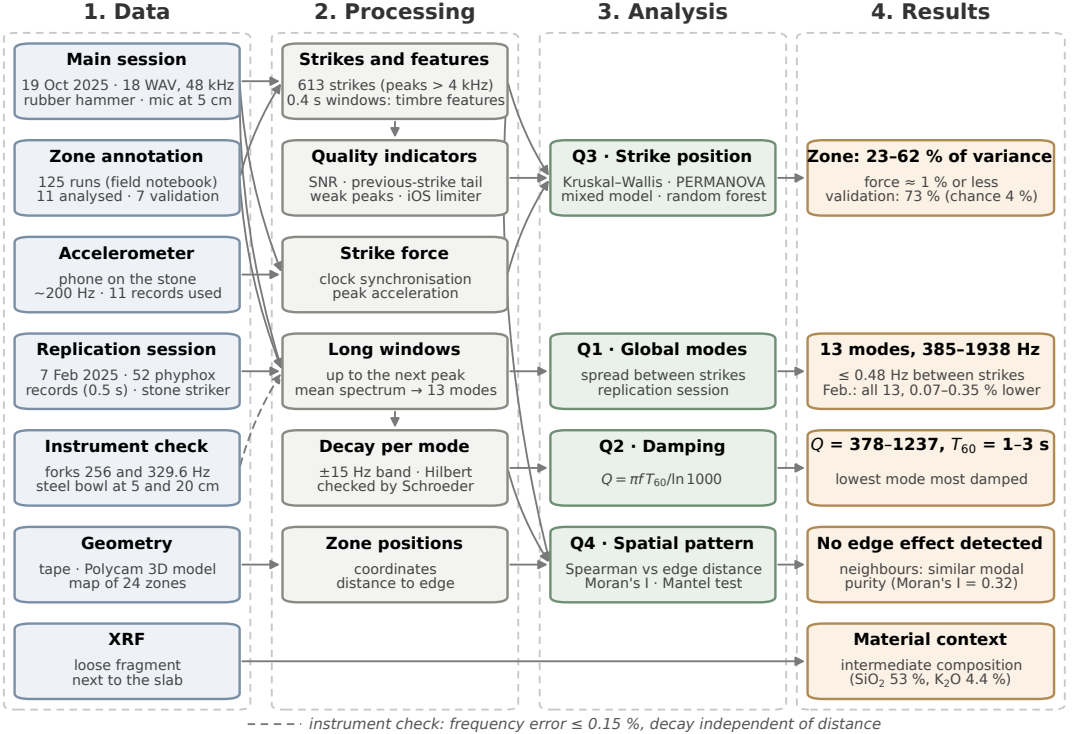
1. Data
2. Processing
3. Analysis
4. Results
Main session
19 Oct 2025 · 18 WAV, 48 kHz
rubber hammer · mic at 5 cm
Zone annotation
125 runs (field notebook)
11 analysed · 7 validation
Accelerometer
phone on the stone
~200 Hz · 11 records used
Replication session
7 Feb 2025 · 52 phyphox
records (0.5 s) · stone striker
Instrument check
forks 256 and 329.6 Hz
steel bowl at 5 and 20 cm
Geometry
tape · Polycam 3D model
map of 24 zones
XRF
loose fragment
next to the slab
Strikes and features
613 strikes (peaks > 4 kHz)
0.4 s windows: timbre features
Quality indicators
SNR · previous-strike tail
weak peaks · iOS limiter
Strike force
clock synchronisation
peak acceleration
Long windows
up to the next peak
mean spectrum → 13 modes
Decay per mode
±15 Hz band · Hilbert
checked by Schroeder
Zone positions
coordinates
distance to edge
Q3 · Strike position
Kruskal–Wallis · PERMANOVA
mixed model · random forest
Q1 · Global modes
spread between strikes
replication session
Q2 · Damping
$Q = \pi f T_{60} / \ln 1000$
Q4 · Spatial pattern
Spearman vs edge distance
Moran's I · Mantel test
Zone: 23–62 % of variance
force ≈ 1 % or less
validation: 73 % (chance 4 %)
13 modes, 385–1938 Hz
≤ 0.48 Hz between strikes
Feb.: all 13, 0.07–0.35 % lower
Q = 378–1237, T_{60} = 1–3 s
lowest mode most damped
No edge effect detected
neighbours: similar modal
purity (Moran's I = 0.32)
Material context
intermediate composition
(SiO_2 53 %, K_2O 4.4 %)
instrument check: frequency error ≤ 0.15 %, decay independent of distance